

$$\begin{aligned} & \frac{1}{16\pi^2} \left(\frac{1}{6} s_\gamma c_\gamma \frac{1}{m_\gamma^2} y_d^{pi2} \overline{y_u}^{si1} (\overline{y_d}^{pr} y_d^{sr} (3 c_\gamma^2 + s_\gamma^2) + \overline{y_u}^{pr} y_u^{sr} (c_\gamma^2 + 3 s_\gamma^2)) - \right. \\ & \frac{1}{18} s_\gamma c_\gamma g_1^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,1,0} [m_1, m_q^p] + \frac{1}{9} s_\gamma c_\gamma g_1^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{3,1,-1} [m_1, m_q^p] - \\ & \frac{1}{18} s_\gamma c_\gamma g_1^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{4,1,-2} [m_1, m_q^p] - \frac{3}{2} s_\gamma c_\gamma g_2^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,1,0} [m_2, m_q^p] + \\ & 3 s_\gamma c_\gamma g_2^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{3,1,-1} [m_2, m_q^p] - \frac{3}{2} s_\gamma c_\gamma g_2^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{4,1,-2} [m_2, m_q^p] - \\ & \frac{8}{3} s_\gamma c_\gamma g_3^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,1,0} [m_3, m_q^p] + \frac{16}{3} s_\gamma c_\gamma g_3^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{3,1,-1} [m_3, m_q^p] - \\ & \frac{8}{3} s_\gamma c_\gamma g_3^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{4,1,-2} [m_3, m_q^p] - s_\gamma c_\gamma \overline{y_d}^{pr} y_d^{pi2} y_d^{sr} \overline{y_u}^{si1} LF_{2,1,0} [\tilde{\mu}, m_d^r] + \\ & 2 s_\gamma c_\gamma \overline{y_d}^{pr} y_d^{pi2} y_d^{sr} \overline{y_u}^{si1} LF_{3,1,-1} [\tilde{\mu}, m_d^r] - \\ & s_\gamma c_\gamma \overline{y_d}^{pr} y_d^{pi2} y_d^{sr} \overline{y_u}^{si1} LF_{4,1,-2} [\tilde{\mu}, m_d^r] - s_\gamma c_\gamma y_d^{pi2} \overline{y_u}^{pr} \overline{y_u}^{si1} y_u^{sr} LF_{2,1,0} [\tilde{\mu}, m_u^r] + \\ & 2 s_\gamma c_\gamma y_d^{pi2} \overline{y_u}^{pr} \overline{y_u}^{si1} y_u^{sr} LF_{3,1,-1} [\tilde{\mu}, m_u^r] - s_\gamma c_\gamma y_d^{pi2} \overline{y_u}^{pr} \overline{y_u}^{si1} y_u^{sr} LF_{4,1,-2} [\tilde{\mu}, m_u^r] + \\ & \frac{1}{18} m_1 s_\gamma g_1^2 \overline{y_u}^{pi1} (-c_\gamma a_d^{pi2} + s_\gamma \tilde{\mu} y_d^{pi2}) LF_{2,1,1,0} [m_1, m_d^{i2}, m_q^p] + \\ & \frac{1}{36} m_1 s_\gamma g_1^2 \overline{y_u}^{pi1} (c_\gamma a_d^{pi2} - s_\gamma \tilde{\mu} y_d^{pi2}) LF_{2,2,1,-1} [m_1, m_d^{i2}, m_q^p] + \\ & \frac{1}{18} m_1 s_\gamma g_1^2 \overline{y_u}^{pi1} (c_\gamma a_d^{pi2} - s_\gamma \tilde{\mu} y_d^{pi2}) LF_{3,1,1,-1} [m_1, m_d^{i2}, m_q^p] + \\ & \frac{1}{36} m_1 s_\gamma g_1^2 \overline{y_u}^{pi1} (-c_\gamma a_d^{pi2} + s_\gamma \tilde{\mu} y_d^{pi2}) LF_{2,2,1,-1} [m_1, m_q^p, m_d^{i2}] + \\ & \frac{1}{9} m_1 c_\gamma g_1^2 y_d^{pi2} (s_\gamma \overline{a_u}^{pi1} - \tilde{\mu} c_\gamma \overline{y_u}^{pi1}) LF_{2,1,1,0} [m_1, m_q^p, m_u^{i1}] + \\ & \frac{1}{18} m_1 c_\gamma g_1^2 y_d^{pi2} (s_\gamma \overline{a_u}^{pi1} - \tilde{\mu} c_\gamma \overline{y_u}^{pi1}) LF_{2,2,1,-1} [m_1, m_q^p, m_u^{i1}] + \\ & \frac{1}{9} m_1 c_\gamma g_1^2 y_d^{pi2} (-s_\gamma \overline{a_u}^{pi1} + \tilde{\mu} c_\gamma \overline{y_u}^{pi1}) LF_{3,1,1,-1} [m_1, m_q^p, m_u^{i1}] + \\ & \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pi1} (-c_\gamma^2 + s_\gamma^2) LF_{2,1,1,0} [m_1, m_q^p, \tilde{\mu}] + \\ & \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pi1} (c_\gamma^2 - s_\gamma^2) LF_{3,1,1,-1} [m_1, m_q^p, \tilde{\mu}] + \\ & \frac{1}{18} m_1 c_\gamma g_1^2 y_d^{pi2} (-s_\gamma \overline{a_u}^{pi1} + \tilde{\mu} c_\gamma \overline{y_u}^{pi1}) LF_{2,2,1,-1} [m_1, m_u^{i1}, m_q^p] - \\ & \frac{1}{6} s_\gamma c_\gamma g_1^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,2,1,-2} [m_1, \tilde{\mu}, m_d^{i2}] - \\ & \frac{1}{6} m_1 \tilde{\mu} g_1^2 s_\gamma^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,2,1,-1} [m_1, \tilde{\mu}, m_d^{i2}] + \frac{1}{4} s_\gamma c_\gamma g_1^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,2,1,-2} [m_1, \tilde{\mu}, m_q^p] + \\ & \frac{1}{12} \tilde{\mu} g_1^2 y_d^{pi2} \overline{y_u}^{pi1} (2 m_1 (2 c_\gamma^2 + s_\gamma^2) + 3 s_\gamma \tilde{\mu} c_\gamma) LF_{2,2,1,-1} [m_1, \tilde{\mu}, m_q^p] - \\ & \frac{1}{3} s_\gamma c_\gamma g_1^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,2,1,-2} [m_1, \tilde{\mu}, m_u^{i1}] - \\ & \frac{1}{3} m_1 \tilde{\mu} g_1^2 c_\gamma^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,2,1,-1} [m_1, \tilde{\mu}, m_u^{i1}] + \frac{9}{2} s_\gamma c_\gamma g_2^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,1,1,-1} [m_2, m_q^p, \tilde{\mu}] + \\ & \frac{3}{2} m_2 \tilde{\mu} g_2^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,1,1,0} [m_2, m_q^p, \tilde{\mu}] - 3 s_\gamma c_\gamma g_2^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{3,1,1,-2} [m_2, m_q^p, \tilde{\mu}] - \\ & \frac{3}{2} m_2 \tilde{\mu} g_2^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{3,1,1,-1} [m_2, m_q^p, \tilde{\mu}] - \frac{9}{4} s_\gamma c_\gamma g_2^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,2,1,-2} [m_2, \tilde{\mu}, m_q^p] - \\ & \frac{3}{4} \tilde{\mu} g_2^2 y_d^{pi2} \overline{y_u}^{pi1} (2 m_2 + s_\gamma \tilde{\mu} c_\gamma) LF_{2,2,1,-1} [m_2, \tilde{\mu}, m_q^p] + \\ & \frac{4}{3} m_3 s_\gamma g_3^2 \overline{y_u}^{pi1} (c_\gamma a_d^{pi2} - s_\gamma \tilde{\mu} y_d^{pi2}) LF_{2,1,1,0} [m_3, m_d^{i2}, m_q^p] + \\ & \frac{2}{3} m_3 s_\gamma g_3^2 \overline{y_u}^{pi1} (-c_\gamma a_d^{pi2} + s_\gamma \tilde{\mu} y_d^{pi2}) LF_{2,2,1,-1} [m_3, m_d^{i2}, m_q^p] + \\ & \frac{4}{3} m_3 s_\gamma g_3^2 \overline{y_u}^{pi1} (-c_\gamma a_d^{pi2} + s_\gamma \tilde{\mu} y_d^{pi2}) LF_{3,1,1,-1} [m_3, m_d^{i2}, m_q^p] + \\ & \frac{2}{3} m_3 s_\gamma g_3^2 \overline{y_u}^{pi1} (c_\gamma a_d^{pi2} - s_\gamma \tilde{\mu} y_d^{pi2}) LF_{2,2,1,-1} [m_3, m_q^p, m_d^{i2}] + \\ & \frac{4}{3} m_3 c_\gamma g_3^2 y_d^{pi2} (s_\gamma \overline{a_u}^{pi1} - \tilde{\mu} c_\gamma \overline{y_u}^{pi1}) LF_{2,1,1,0} [m_3, m_q^p, m_u^{i1}] + \\ & \frac{2}{3} m_3 c_\gamma g_3^2 y_d^{pi2} (s_\gamma \overline{a_u}^{pi1} - \tilde{\mu} c_\gamma \overline{y_u}^{pi1}) LF_{2,2,1,-1} [m_3, m_q^p, m_u^{i1}] + \\ & \frac{4}{3} m_3 c_\gamma g_3^2 y_d^{pi2} (-s_\gamma \overline{a_u}^{pi1} + \tilde{\mu} c_\gamma \overline{y_u}^{pi1}) LF_{3,1,1,-1} [m_3, m_q^p, m_u^{i1}] + \\ & \frac{2}{3} m_3 c_\gamma g_3^2 y_d^{pi2} (-s_\gamma \overline{a_u}^{pi1} + \tilde{\mu} c_\gamma \overline{y_u}^{pi1}) LF_{2,2,1,-1} [m_3, m_u^{i1}, m_q^p] + \\ & \frac{1}{4} \tilde{\mu} c_\gamma \overline{y_d}^{pr} y_d^{pi2} \overline{y_u}^{si1} (-c_\gamma a_d^{sr} + s_\gamma \tilde{\mu} y_d^{sr}) LF_{2,2,1,-1} [m_d^r, \tilde{\mu}, m_q^s] + \\ & \frac{1}{4} s_\gamma \tilde{\mu} y_d^{pi2} \overline{y_u}^{si1} y_u^{sr} (s_\gamma \overline{a_u}^{pr} - \tilde{\mu} c_\gamma \overline{y_u}^{pr}) LF_{2,2,1,-1} [m_q^p, \tilde{\mu}, m_u^r] + \\ & \frac{1}{4} \tilde{\mu} c_\gamma \overline{y_d}^{pr} y_d^{pi2} \overline{y_u}^{si1} (c_\gamma a_d^{sr} - s_\gamma \tilde{\mu} y_d^{sr}) LF_{2,2,1,-1} [m_q^s, \tilde{\mu}, m_d^r] + \\ & \frac{1}{4} s_\gamma \tilde{\mu} y_d^{pi2} \overline{y_u}^{si1} y_u^{sr} (-s_\gamma \overline{a_u}^{pr} + \tilde{\mu} c_\gamma \overline{y_u}^{pr}) LF_{2,2,1,-1} [m_u^r, \tilde{\mu}, m_q^p] + \\ & \frac{1}{2} s_\gamma c_\gamma g_1^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,1,1,-1} [\tilde{\mu}, m_1, m_d^{i2}] + \frac{1}{3} m_1 \tilde{\mu} g_1^2 s_\gamma^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,1,1,0} [\tilde{\mu}, m_1, m_d^{i2}] - \\ & \frac{1}{3} s_\gamma c_\gamma g_1^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{3,1,1,-2} [\tilde{\mu}, m_1, m_d^{i2}] - \frac{1}{3} m_1 \tilde{\mu} g_1^2 s_\gamma^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{3,1,$$